

Library Management System – Spring Security OAuth2 Hands-on

Objective

Secure a Library Management System using:

- Spring Authorization Server
 - Spring OAuth2 Resource Server
 - JWT Authentication
 - Spring Security
 - BCrypt Password Encoder
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Applications

1. Authorization Server

Responsibilities:

- Authenticate users
- Store users in H2 database
- Encode passwords using BCryptPasswordEncoder
- Issue JWT access tokens
- Register OAuth2 clients

2. Resource Server

Responsibilities:

- Validate JWT access tokens
 - Secure REST APIs
 - Authorize users based on roles
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Users (H2 Database)

Username	Password	Role
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member	member123	MEMBER
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librarian	librarian123	LIBRARIAN
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admin	admin123	ADMIN
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Store passwords **encoded**, not as plain text.

Part 1 – Client Credentials Flow

Configure an OAuth2 client that uses the **Client Credentials** grant.

Tasks

- Register an OAuth2 client.
- Request an access token.
- Call GET /api/books using the token.
- Verify that the Resource Server validates the JWT.

Part 2 – Authorization Code Flow

Configure an OAuth2 client that uses the **Authorization Code** grant.

Tasks

- Authenticate users from the H2 database.
- Encode passwords using BCryptPasswordEncoder.
- Generate JWT tokens after successful login.

Expected Access

MEMBER

- ☒ List books
- ☒ Borrow books
- ☒ Return books

LIBRARIAN

- ☒ List books
- ☐ Borrow books
- ☐ Return books

ADMIN

- ☒ Access all books CRUD endpoints
- ☐ Borrow books
- ☐ Return books

Deliverables

- Authorization Server project
- Resource Server project
- Users stored in H2
- Passwords encoded with BCrypt
- Working Client Credentials flow
- Working Authorization Code flow